

NOSHIN FOUZIA TASNIM

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[LinkedIn](#) || [Github](#) || [Portfolio](#)

RESEARCH INTEREST

Machine Learning, Image Processing, Biomedical Image Analysis, Human Computer Interaction, Large Language Models

WORK EXPERIENCE

Intern, Product Management, App & Essential Services Jan 2026 - Present

Product & Technology Division, bKash Ltd

- Worked in developing the Source of Truth for bKash's virtual assistant, AVA's deployment
- Contributed to dataset preparation and schema for new AVA conversational flow
- Researched on LLMs, RAG, annotation practices
- Customer Care Center visit to analyse demands of consumers regarding bKash services

Undergraduate Teaching Assistant/ Student Tutor Jun 2024 - May 2025

CSE Department, BRAC University

- Assigned to Algorithms course and helped 100+ students during lab classes for assigned section
- Checked 200+ lab quiz scripts, assignments and mentored students offering 12 hours consultation each week

Executive

BASIS Students Forum BRACU Chapter

Jan 2024 - Dec 2024

- Coordinated between BASIS and BUCC to organize different events and seminars
- Contributed in meetings and planning for NASA Space Apps Challenge 2024

EDUCATION

Master of Science in Computer Science and Engineering Jan 2026 - Present

BRAC University, Dhaka

Bachelor of Science in Computer Science and Engineering Oct 2021 - Oct 2025

BRAC University, Dhaka

Credits Completed: 136
CGPA: 3.93/4.00

- Notable Courses: Data Structure & Algorithms, Artificial Intelligence, Data Science, Computer Networks, Computer Architecture, Microprocessors, Software Engineering, Database Systems

Higher Secondary School Certificate (Science) July 2018 - Jan 2021

Viqarunnisa Noon School and College, Dhaka

GPA: 5.00/5.00

Secondary School Certificate (Science) Jan 2008 - Apr 2018

Viqarunnisa Noon School and College, Dhaka

GPA: 5.00/5.00

RESEARCH EXPERIENCE

Classification of Alzheimer's and Dementia subtypes using R-STDP Driven Spiking Neural Networks

AI/ML, Neuroimaging, Undergraduate Thesis

[Repository](#)

Co-developed an SNN autoencoder model for classification of MRI images of demented and non-demented patients under the supervision of my thesis supervisor, Dr. Md. Golam Rabiul Alam and co supervisor, Zayed Humayun. We built a primary dataset by collecting brain MRI images from the National Institute of Neurosciences for 8 months with IRB approval. My contributions included data collection, MRI image preprocessing, building the model and hyperparameter tuning for the best results. The model incorporates Forward-Forward Learning and R-STDP for training without backpropagation and ensemble classifier for testing. The model achieved a 77.92% accuracy and 79.47% precision on demented vs non-demented classification. The model was optimized using a CNN branch for feature extraction which achieved a 86.98% accuracy.

- Currently preparing the paper for publication in IEEE Journal of Biomedical and Health Informatics

Usability assessment of e-Tax service platform in Bangladesh across different user groups

Human Computer Interaction, Ongoing Research

Conducting research for assessing the usability of e-Return platform for submitting tax return among different user groups in Bangladesh based on age and profession under the supervision of Dr. Farida Chowdhury. This is an ongoing research and we are working to evaluate the usability of this platform using different metrics. The data collection process includes interviewing participants from different user groups including developers, documenting and analyzing user experience and feedback regarding specific task flows of the platform. The user experiences' data will be used to evaluate the usability of the platform.

TECHNICAL SKILLS

- **Languages:** Python, C/C++, Java, Javascript, PHP, SQL, HTML, CSS, RISC-V, x86 assembly language
 - **Frameworks & Libraries :** Laravel, Next.js, PyTorch, Keras, Matplotlib, TensorFlow, Pandas, Scikit-learn
 - **Development Tools:** LaTeX, Git, Github, Postman, Jira, Flex, Bison
 - **Database:** mySQL, PostgreSQL
 - **Hardware design and Networking Tools:** LTSpice, Quartus, Verilog HDL, Cisco Packet Tracer, Wireshark
 - **Microcontrollers:** STM32, Arduino, RaspberryPi
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ACADEMIC PROJECTS

Club Management System

A dynamic website created for club management with deployment

[Repository](#)

- There were different panels for admin, user and clubs with club registration, roles, events, email and notification system
- Next.js framework used, PostgreSQL used for database and Vercel for deployment

Credit Card Fraud Detection

A machine learning project made for detection of fraud credit cards

[Repository](#)

- The dataset was oversampled and undersampled to reduce class imbalance and preprocessed
- Machine learning models KNN, Logistic Regression, Decision Tree & random forest were trained and tested

Smart Water Cleaner Robot

A robot built using Arduino for collecting trash from water and monitoring water quality

[Repository](#)

- Arduino Uno R3 used as microcontroller, C used for arduino code and L293D motor driver shield used for driving motors
- The robot can navigate using HC-SR04, collect trash and monitor water quality using temperature and turbidity sensor and display it on an I2C LCD

Travelcom

A website created for finding travel partner and planning trips

[Repository](#)

- Laravel framework used for backend, HTML, CSS, React used for Frontend
 - Admin notification feature, user chatting and group trip features implemented
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EXTRACURRICULARS

Assistant Director

Oct 2023 - Dec 2024

Event Management Department, BRAC University Computer Club (BUCC)

- Successfully organized 25+ events, workshops and webinars and led volunteer teams collaborating with other departments
- Managed permissions and received guests and event sponsors; experienced in food allotment and refreshment
- Created a guideline book on venue booking rules in accordance with university policies
- Planned and supervised activities of BUCC Study Corner

Organizer, Intrahacktive 1.0

Oct 2024 - Nov 2024

Event Management Department, BRAC University Computer Club (BUCC)

- Helped with planning and execution of the entire event and organized hackathon, database wizard, debugger
- Led the logistics team and worked with volunteers, guest reception

Member, IEEE Computer Society

Apr 2022 - Dec 2022

IEEE BRAC University Student Branch

- Participated in different workshops on AI/ML and Cybersecurity
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HONORS AND AWARDS

• Vice Chancellor's List

For obtaining a GPA of 3.9+ for 9 semesters

• Dean's List

For obtaining a GPA of 3.7+

• Scholarship based on previous results

Received 50% waiver on tuition fees at BRAC University for maintaining a 3.8+ CGPA

• Bronze Award at Duke of Edinburgh International Awards

It was a 6-month long program combining skill development, physical activity, voluntary service and adventurous journey

REFERENCE

Dr. Md. Golam Rabiul Alam

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Rafeed Rahman

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